

Teacher Guide — middle / module-07-eleven-multiplication

IKS Curriculum

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Overview

Module 7 — Multiplication by 11 (*Ekādhikena Pūrveṇa*)

Age band: Middle · **Grades:** 6–8 · **Duration:** 10 sessions × 45 min

Learning objectives

By the end of this module, students will be able to:

1. **Execute** the $\times 11$ “split-and-add” trick on any 2-digit number, with and without the carry case, in under 3 seconds per problem.
2. **Extend** the trick to 3-digit numbers using the running “neighbour-sum” pattern (e.g. $234 \times 11 \rightarrow 2_5_7_4 \rightarrow 2574$).
3. **Prove** algebraically why the trick works: $(10a + b) \times 11 = 100a + 10(a + b) + b$.
4. **Apply** the *Anurūpyeṇa* extension to multiply 2-digit numbers by 12, 13, ..., 19.
5. **Compare** the $\times 11$ shortcut against standard long multiplication and identify when each is faster.
6. **Locate** the trick honestly in the history of Indian mathematics — as Bharati Krishna Tirthaji’s 1965 reformulation, not a verifiable Vedic teaching.

NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems (Vedic-math sūtras) with mainstream arithmetic.
- **§4.23** — Experiential learning via timed mental-math races and paired drills.
- **§4.27** — Local context: Tirthaji’s Indian mathematical-pedagogy tradition.
- **§11.8** — Holistic, multidisciplinary education (algebra $\leftrightarrow$ mental arithmetic $\leftrightarrow$ history of science).

Prerequisites

- Confident with addition of two single-digit numbers up to 18 (carry handling).
- Comfortable with standard long multiplication of 2-digit $\times$ 1-digit numbers.
- Knows what a variable is and can read expressions like $10a + b$.

Key vocabulary

See `_shared/glossary.md`.

- *Ekādhikena Pūrveṇa* — “by one more than the previous” (1st of Tirthaji’s 16 sūtras)
- *Anurūpyeṇa* — “proportionally” (sub-sūtra used to extend $\times 11$ to $\times 12 \dots \times 19$)
- *sūtra* — short formula / aphorism
- *Tirthaji* — Bharati Krishna Tirthaji (1884–1960), author of *Vedic Mathematics* (1965)

Materials checklist

- A4 worksheets (one per student, varied difficulty)
- Stopwatches or phone timers (1 per pair)
- Wall poster: the “neighbour-sum” rule visualised as $2_{(2+3)}_3 = 253$
- Whiteboards or scrap paper for live demos
- Optional: printed extract of Tirthaji’s preface for Day 8

Day-by-day overview

Day	Title	Focus	Assessment
1	The Hook — $2_3 = 253$	Reveal the trick on no-carry 2-digit cases	Self-timed baseline
2	The Carry Case	What happens when the middle digit overflows	Pattern quiz
3	Why It Works	Algebraic proof of $(10a+b)\times 11$	Reflection paragraph
4	Three-Digit $\times 11$	Running-sum extension	Formative quiz A
5	Mental-Math Speed Day	Timed paired race	Formative quiz B
6	<i>Anurūpyeṇa</i> — $\times 12$ to $\times 19$	Scaled-middle extension family	Activity rubric
7	When the Trick Doesn’t Help	Limits; comparison to long multiplication	Reflection
8	Tirthaji’s Sūtra System in Context	Historical framing (honest)	Discussion
9	Mixed Practice + Project Session	Drill + project work	Project draft
10	Final Assessment + Project Share	Summative + 2-min presentations	Summative + project

Citation sources used

The vidya-karana RAG does not cover Vedic-math sources. This module draws from:

- **Tirthaji, Bharati Krishna.** *Vedic Mathematics* (1965), Motilal Banarsidass. Sūtra 1 (*Ekādhikena Pūrveṇa*); sub-sūtra *Anurūpyeṇa*. Cited from the SutraGanita corpus PDF.
- **SutraGanita corpus** (/Users/sree/Projects/SutraGanita/content/) — *Microsoft Word - sutras1.pdf* (sūtra list).
- **Dani, S. G.** “Myths and Reality: On Vedic Mathematics” (1993, reprinted *Resonance* October 2001) — for honest historical framing on Day 8.

Full citation chain in `sources.md`.

Project brief (one-line teaser)

Each pair produces ONE of three artefacts: a 60-second teaching video of the $\times 11$ trick for a Grade 4 audience, a printed workbook of 25 problems across difficulty levels, OR a one-page “proof poster” deriving the algebraic identity. Full brief in `assessments/project-brief.md`.

10-Day Lesson Plan

Module 7 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Hook — $2_3 = 253$	<code>days/day-01-hook.md</code>	Self-timed baseline; reveal
2	The Carry Case	<code>days/day-02-carry-case.md</code>	Pattern quiz
3	Why It Works	<code>days/day-03-why-it-works.md</code>	Reflection paragraph
4	Three-Digit $\times 11$	<code>days/day-04-three-digit.md</code>	<code>quizzes/formative.md</code> Part A
5	Mental-Math Speed Day	<code>days/day-05-speed-day.md</code>	<code>activities/activity-01-speed-race.md</code> + Part B
6	<i>Anurūpyeṇa</i> — $\times 12$ to $\times 19$	<code>days/day-06-anurupyena.md</code>	<code>activities/activity-02-anurupyena.md</code>
7	When the Trick Doesn’t Help	<code>days/day-07-limits.md</code>	Reflection
8	Tirthaji’s Sūtra System in Context	<code>days/day-08-tirthaji-context.md</code>	Discussion

Day	Session title	Lesson file	Activity / assessment
9	Mixed Practice + Project Session	days/day-09-project-session.md	Project draft
10	Final Assessment + Project Share	days/day-10-final-assessment.md	quizzes/summative.md + project

Pacing notes

- Day 1 establishes a personal baseline (how long to compute 73×11 the long way). Day 5 retests with the trick. The visible speedup is the motivation.
- Days 1–3 are the core. If your class is slow on Day 2 (the carry case), spend a second day there before moving on; combine Days 4 and 5 to recover.
- Day 6 (*Anurūpyeṇa* extension to $\times 12 \dots \times 19$) can be lighter or expanded depending on student appetite.
- Day 7 (“when the trick doesn’t help”) is short by design — it’s the honesty hour. $\times 23$ doesn’t fit the family; don’t pretend it does.
- Day 8 is the cultural-context day. Keep it inquiry-driven; don’t lecture.
- Days 9–10 are project-light by design — the project here is smaller-scope than Module 5’s, because the technique is narrower.

Daily warm-up (recurring)

Every class starts with **2 minutes of “neighbour-sum” recall**: teacher calls a 2-digit number; students call back its $\times 11$ product. Standardised set in `homework.md`.

What students leave the module with

- Confident mental $\times 11$ on any 2-digit (and most 3-digit) number, including the carry case.
- One memorised algebraic identity: $(10a + b) \times 11 = 100a + 10(a + b) + b$.
- An extension pattern for $\times 12$ to $\times 19$ using *Anurūpyeṇa*.
- Honest understanding that the trick is Tirthaji’s 20th-century reformulation, not a verifiable ancient Vedic teaching.

Teacher Notes

Teacher Notes — Module 7 Eleven Multiplication (Middle)

Posture

This module is about **mental fluency plus algebraic understanding**. The two reinforce each other: students who only memorise the trick freeze on the carry case; students who only prove the algebra never get the speed. We want both.

The Sanskrit name *Ekādhikena Pūrveṇa* is the cultural anchor. The *trick* is what students take home for life.

Honesty about provenance. The $\times 11$ technique is widely taught as “Vedic mathematics.” The honest framing for this module:

- Tirthaji (1965) names the trick using a Sanskrit sūtra (*Ekādhikena Pūrveṇa*).
- The book was published posthumously (Tirthaji died 1960; book 1965).
- Tirthaji claims the sūtras come from an *Atharva-veda Pariśiṣṭa*, but no other scholar has produced this manuscript.
- The trick works mathematically; the “Vedic” attribution is academically disputed (Dani 1993, Plofker 2009).

At Middle band we state this briefly on Day 8 and let students sit with the ambiguity. Don’t moralise either way.

Common misconceptions

1. **“The middle digit IS the sum, even when it’s two digits.”** Students write 73×11 as 7-10-3 (a “5-digit number” with a 10 in the middle). The carry step fixes this; reinforce Day 2.
2. **“Add 1 to the answer, not to the leftmost digit.”** When the digit-sum carries, the +1 goes into the **first digit**, not into the running answer. Watch for “ $73 \times 11 = 8(10)3$ ” being written as 813.
3. **“It’s just multiplication, so why a special name?”** True observation. Don’t dismiss it. Reframe: the name labels a *pattern*; the pattern lets you do this mentally instead of writing it out. Names matter when they make things fast.
4. **“The trick is Vedic.”** Most students arrive believing this. Day 8 is where we honour their belief but also lay out the academic critique. No shaming.
5. **“3-digit $\times 11$ is too hard to do in my head.”** Almost no one can on Day 4. By Day 5–6 most students can. Practice gap fills slowly.

6. **“Anurūpyeṇa is too hard.”** It is harder than $\times 11$ — that’s structural. Some classes may benefit from skipping $\times 17/\times 19$ and staying with $\times 12/\times 13$.

Differentiation hints

- **Tier up:** Push to 4-digit $\times 11$, then 3-digit $\times 12$ to 19. The brave will love the cascading-carry challenge in 4-digit $\times 11$.
- **Tier down:** Stay on 2-digit $\times 11$ (no carry) until Day 3. Some students need the foundation before any extension.

Differential pacing

- A class with strong arithmetic baseline can compress to 7 days by combining Days 4+5 (3-digit + speed day) and Days 7+8 (limits + history).
- A class with shaky multiplication baseline should NOT skip Day 3 (the proof) — the algebra is what makes the trick reliable rather than memorised-and-forgotten.

Sensitivity notes

- Some students will have parents or coaching backgrounds where the trick is taught as ancient sacred knowledge. Be respectful. Frame the academic critique as “what the scholarly consensus says,” not “what your parents got wrong.”
- Avoid civilisational comparisons (“Indians had this before Westerners”). The technique is its own thing; what makes it Indian is Tirthaji’s synthesis and the Sanskrit naming, not any race for priority.
- The competitive-mental-math kids will be ahead from Day 1. Use them as peer-coaches; reframe their advantage as “you’ll explain the WHY this week.”

Project oversight

- Three formats (video, workbook, proof poster) each have different challenges:
 - **Video:** check before Day 9 that the pair has phone access and a quiet recording space.
 - **Workbook:** simple to start, easy to abandon. Set a 15-problem minimum.
 - **Proof poster:** recommended for pairs who struggled on Day 3 — making the algebra public forces internalisation.
- The project here is intentionally smaller than Module 5’s. Don’t let it bloat.

Citation discipline

- Citations come from the SutraGanita PDF corpus, plus Tirthaji (1965) and Dani (1993/2001).
- Students should cite specific PDFs or books in their projects. The standard is <source>, <section/page>.

If you only have 6 days

- Day 1: Hook + baseline
- Day 2: Carry case

- Day 3: Why it works (algebra)
- Day 4: Combined — $3\text{-digit} \times 11 + \textit{Anurūpyeṇa}$ intro
- Day 5: Combined — limits + Tirthaji context
- Day 6: Final assessment, no project

Skip the project entirely if needed; keep the algebra.

What I wish someone had told me first time teaching this

- The cascading-carry case ($99 \times 11 = 1089$) is dramatic. Use it on Day 2 and milk it.
- Students who can do the trick fast but can't explain why are NOT mastered yet. Push them to verbalise.
- The Day 8 discussion can feel deflating if you're not careful. Frame Tirthaji as a real intellectual achievement BEFORE introducing Dani's critique. "Both can be true."

Sources

Sources — Module 7 Eleven Multiplication (Middle)

The **vidya-karana RAG** does not cover Vedic-math sources, so this module draws directly from a small set of published references plus the SutraGanita PDF corpus (`/Users/sree/Projects/SutraGanita/content/`).

Primary source

Tirthaji, Bharati Krishna. *Vedic Mathematics: Sixteen Simple Mathematical Formulae from the Vedas* (1965, edited posthumously by V. S. Agrawala). Delhi: Motilal Banarsidass.

- **Sūtra 1 — *Ekādhikena Pūrveṇa*** ("by one more than the previous"). Tirthaji applies this sūtra mainly to recurring-decimal division (e.g. $1/19 = 0.052631\dots$); the $\times 11$ multiplication trick is presented as a related corollary. The version we teach in this module is the standard pedagogical form found in every Vedic-math secondary reference.
- **Sub-sūtra — *Anurūpyeṇa*** ("proportionally"). Used to scale the $\times 11$ pattern to $\times 12$, $\times 13$, ..., $\times 19$.
- The book was completed in manuscript form between 1911 and 1918 (per the editor's foreword) and published in 1965, five years after Tirthaji's death (1960).

Cite as: Tirthaji (1965), *Vedic Mathematics*, sūtra 1.

Supporting sources (SutraGanita corpus)

- **Microsoft Word - sutras1.pdf** — single-page reference listing all 16 main sūtras and 13 sub-sūtras with English glosses. Used on Day 8 ("Tirthaji's Sūtra System in Context").
- **Vedic_Maths.pdf** (ISKCON Desire Tree) — workbook-style. Contains practice sets for the $\times 11$ trick. Used for Day 4–5 drills.

- **Vedic Mathematics Batch1.pdf** (Vedic Cultural Center, Sammamish WA) — secondary reference with worked examples of the $\times 11$ method, including the carry case.

Historical critique (used on Day 8)

- **Dani, S. G.** “Myths and Reality: On Vedic Mathematics.” *Frontline* (1993), reprinted in *Resonance* vol. 6 no. 10, October 2001, pp. 53–65. Argues that no manuscript of the *Atharva-veda Pariśiṣṭa* containing these 16 sūtras has been produced, and that the techniques are best understood as a 20th-century mental-arithmetic synthesis.
- **Plofker, Kim.** *Mathematics in India* (2009), Princeton University Press. Standard reference on the documented history of pre-modern Indian mathematics; covers (e.g.) Brahmagupta, Bhāskara II, the Kerala school. Tirthaji’s sūtras are mentioned in passing as a 20th-century claim outside the documented manuscript tradition.

What we cite to students at Middle band

Per the citation rule in `_shared/style-guide.md`, Middle band needs at least one paraphrase with citation per module. We use:

Tirthaji (1965) names the underlying sūtra *Ekādhikena Pūrveṇa* — “by one more than the previous.”

...on Day 1 and again on Day 8 with fuller context.

What we DON’T do

We do not claim the $\times 11$ trick “comes from the Vedas.” We say it is presented in Tirthaji’s 1965 book under a Sanskrit sūtra name; whether the sūtra itself is verifiably Vedic is the question raised honestly on Day 8.

Day 1 — The Hook: $2_3 = 253$

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will compute any 2-digit $\times 11$ product where the digit-sum is less than 10, in under 3 seconds, and will have recorded a personal baseline time for the same problem using standard long multiplication.

Materials

- Printed baseline sheet: ONE multiplication problem (73×11). One per student.
- Stopwatch or wall clock with a second hand. (One per pair is plenty.)
- Whiteboard / large paper to demonstrate.

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: ekādhikena pūrveṇa; lit. “by one more than the previous”) — the Sanskrit name Tirthaji (1965) gives the underlying sūtra. We use it as the cultural anchor; the technique is what students take home.

Lesson flow

Warm-up (5 min)

- Teacher: “How long does it take you to compute 73×11 the normal way? Make a guess.”
- Take 3–4 guesses. Write the range on the board.

Baseline timing (8 min)

- Hand out the baseline sheet face-down. Students write their name and predicted time on the back.
- On “Go,” flip and solve 73×11 using long multiplication.
- Record actual time honestly. Mark answer themselves.
- Expected range: 15–60 seconds.

Teacher demonstration (12 min) — The trick

- Write on the board:

$$73 \times 11 = ?$$

- “I’m going to show you a method that takes about 2 seconds, then we’ll figure out why it works.”

Step 1: Write the digits, with a gap: 7 _ 3

Step 2: Add the two digits: $7 + 3 = 10$.

Wait – that’s the carry case. Park it.

- *Reset. Different number first. Try 23×11 :*

Step 1: Split: 2 _ 3

Step 2: Add: 2 + 3 = 5

Step 3: Drop in: 2 5 3 → 253.

Check on the board with long multiplication: $23 \times 11 = 23 \times 10 + 23 = 230 + 23 = 253$. ✓

- Try 45×11 :

4 _ 5 → 4 + 5 = 9 → 4 9 5 → 495.

- Try 62×11 :

6 _ 2 → 6 + 2 = 8 → 6 8 2 → 682.

- “Pattern: outer digits stay the same; middle digit = their sum.”
- Now name it. “Tirthaji (1965) calls this **Ekādhikena Pūrveṇa** — ‘by one more than the previous.’ That is the Sanskrit name of the sūtra. The trick itself is just $(10a + b) \times 11 = 100a + 10(a+b) + b$, which we’ll prove on Day 3.”

Activity (15 min) — Try it yourself

- Worksheet (handed out): 10 two-digit $\times 11$ problems, ALL with digit-sum < 10 .

$$\begin{array}{ccccc} 12 \times 11 & 23 \times 11 & 34 \times 11 & 45 \times 11 & 52 \times 11 \\ 61 \times 11 & 16 \times 11 & 27 \times 11 & 35 \times 11 & 44 \times 11 \end{array}$$

- Pair work. Partner times each other. Goal: each problem under 3 seconds.
- After 10 min, students re-do 73×11 on the back of the baseline sheet. *Hmm — $7 + 3 = 10$. That’s tomorrow’s lesson.*

Wrap-up (5 min)

- Show of hands: who finished all 10 in under 30 seconds? (Most will.)
- Tease tomorrow: “ 73×11 broke the rule because $7 + 3 = 10$ and you can’t write 10 in the middle of a number. Tomorrow — the carry case.”

Student-facing content

The $\times 11$ Trick (no-carry case)

For any 2-digit number with digit-sum < 10 : 1. **Split** the digits with a gap. 2. **Add** the two digits. 3. **Drop** the sum into the gap.

Example: $34 \times 11 \rightarrow 3_4 \rightarrow 3+4=7 \rightarrow 374$.

Tirthaji (1965) names the underlying sūtra **Ekādhikena Pūrveṇa** — “by one more than the previous.” The trick is older than its modern Sanskrit name; the algebra (which we’ll do on Day 3) is the same in every language.

Homework

- Find any two 2-digit numbers (digit-sum < 10) in your house — phone numbers, page numbers, prices. Compute each $\times 11$ using the trick. Bring 5 worked examples to class tomorrow.

Differentiation

- **Tier up:** Try 3-digit $\times 11$ (e.g. 234×11 , 152×11). See if you can guess the pattern before Day 4.
- **Tier down:** If the gap is confusing, write the answer as a 3-digit number directly: hundreds = first digit, tens = sum, units = second digit.

Teacher notes

- **The baseline time matters.** Day 5’s retest depends on having a Day 1 record.

- Don't dwell on the Sanskrit name today. *Ekādhikena Pūrveṇa* is the cultural anchor; the win is mental speed.
- Some students will already know the trick (especially those from competitive-math backgrounds). Have them be peer-coaches. Reframe their advantage as “you'll learn the WHY this week.”
- About 30% of randomly-chosen 2-digit numbers have digit-sum ≥ 10 (the carry case). So today's worksheet hand-picks the no-carry ones; tomorrow we add the carries.

Citation(s) used in this lesson

- Tirthaji, B. K. (1965). *Vedic Mathematics*. Motilal Banarsidass. Sūtra 1, *Ekādhikena Pūrveṇa*.

Day 2 — The Carry Case

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will correctly handle the carry case in 2-digit $\times 11$ (where the digit sum ≥ 10), including problems like $87 \times 11 = 957$, in under 5 seconds.

Materials

- Whiteboard.
- Printed worksheet of 12 problems, mixing no-carry and carry cases.
- Stopwatch.

Lesson flow

Warm-up (5 min)

- Neighbour-sum drill: teacher calls a 2-digit number with digit-sum < 10 ; class chorus the $\times 11$ product.
- “12 times 11!” $\rightarrow$ “132!” — “34 times 11!” $\rightarrow$ “374!” — 10 in a minute.

Recall (3 min)

- Yesterday's problem: 73×11 . $7 + 3 = 10$. Why was this awkward?
- “You can't put 10 in the middle of a 3-digit number. So what do we do? Carry the 1 — exactly like in standard addition.”

Core (15 min) — The carry case

1. Demonstrate 73×11 fully:

Step 1: Split: 7 3

Step 2: Add: 7 + 3 = 10

Step 3: Write the units digit of the sum (0) in the middle: 7
0 3

Step 4: Carry the tens digit of the sum (1) into the leftmost digit: $7 + 1 = 8$.

Step 5: Answer: $8 \ 0 \ 3 \rightarrow 803$.

Check: $73 \times 11 = 73 \times 10 + 73 = 730 + 73 = \mathbf{803}$. ✓

2. **Try 87×11 :**

$8 \quad \underline{\quad} \quad 7$

$8 + 7 = 15$

Middle = 5; carry 1 into the 8 $\rightarrow$ 9.

Answer: $9 \ 5 \ 7 \rightarrow 957$.

Check: $87 \times 11 = 870 + 87 = \mathbf{957}$. ✓

3. **Try 99×11 :**

$9 \quad \underline{\quad} \quad 9$

$9 + 9 = 18$

Middle = 8; carry 1 into the 9 $\rightarrow$ 10.

But 10 isn't a single digit – so it becomes a "1" carried further: leftmost = 0, with a 1 carried to the left of that.

Answer: $1 \ 0 \ 8 \ 9 \rightarrow 1089$.

Check: $99 \times 11 = 990 + 99 = \mathbf{1089}$. ✓ (4-digit answer because the carry cascaded.)

4. **The full procedure** (boxed on the board):

For any 2-digit number $ab \times 11$:

1. Compute $a + b$.
2. If the sum is < 10 : answer is $a (a+b) b$.
3. If the sum is ≥ 10 : middle digit = units digit of sum; add 1 to a .
4. If the new a is ≥ 10 (only happens at 99): cascade the carry.

Activity (17 min) — Mixed practice

- Worksheet of 12 problems, half no-carry, half carry:

21×11	35×11	47×11	58×11
62×11	74×11	83×11	89×11
91×11	99×11	18×11	27×11

- Round 1 (8 min): solo, untimed. Self-check with long multiplication on any 3 of them.
- Round 2 (8 min): paired. Partner calls a problem; you answer aloud. Goal: under 5 seconds per problem.

Wrap-up (5 min)

- Quick chorus: “What does the trick break into?” — Split, add, drop (and carry if needed).
- Tease Day 3: “Tomorrow we *PROVE* why this works. You’ll write the algebra yourself.”

Student-facing content

The $\times 11$ Trick (with carry)

1. Split the digits.
2. Add them.
3. **If the sum < 10 :** drop it in the middle. Done.
4. **If the sum ≥ 10 :** keep the units digit of the sum as the middle; add 1 to the leftmost digit.

Examples: $- 45 \times 11: 4+5=9 \rightarrow 495 = \mathbf{495}$ - $87 \times 11: 8+7=15 \rightarrow 8(\mathbf{1})57 \rightarrow 957 = \mathbf{957}$ -
 $99 \times 11: 9+9=18 \rightarrow 9(\mathbf{1})89 \rightarrow \text{cascade} \rightarrow 1089 = \mathbf{1089}$

Homework

- 8 mental $\times 11$ problems (carry case): 76, 84, 85, 88, 93, 96, 67, 78 — multiplied by 11. Write the answers.
- Time yourself. Goal: under 30 seconds total.

Differentiation

- **Tier up:** Find a 2-digit number where the trick gives a 4-digit answer (i.e. $99 \times 11 = 1089$). Can you find any others, or is 99 unique? Justify.
- **Tier down:** Solve only the no-carry problems on the worksheet today. We'll cement carries on Day 4.

Teacher notes

- The most common error: students remember to write the units digit of the middle sum but **forget** to add the carry into the leftmost digit. Watch for “ $73 \times 11 = 703$ ” with no carry added — this is a classic mistake.
- The 99×11 case (and the cascading carry) genuinely surprises students. Lean into it — it's evidence the algebra is doing real work, not arithmetic by coincidence.
- If a student asks “why does this work?” — that's Day 3. Park it. Honour the curiosity by writing the question on a board sticky.

Citation(s) used in this lesson

- Tirthaji (1965), sūtra 1 (*Ekādhikena Pūrveṇa*) — the technique name.
- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center) — worked examples of the carry case.

Day 3 — Why It Works

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will derive the identity $(10a + b) \times 11 = 100a + 10(a + b) + b$ and explain in their own words why the $\times 11$ trick is not a coincidence.

Materials

- Whiteboard.
- Worksheet: “explain to a friend” prompt with space for student algebra.

Lesson flow

Warm-up (5 min)

- Neighbour-sum chorus: 15 quick $\times 11$ problems, mix of carry and no-carry.
- “By now your hands know the trick. Today your brain catches up.”

Core (20 min) — The algebra

1. **Write any 2-digit number as $10a + b$.** Make sure every student gets this.

$$23 = 10 \cdot 2 + 3 \quad \rightarrow \quad a = 2, \quad b = 3$$

$$87 = 10 \cdot 8 + 7 \quad \rightarrow \quad a = 8, \quad b = 7$$

2. **Distribute the multiplication by 11.** Step by step on the board, students copy:

$$\begin{aligned}(10a + b) \times 11 &= (10a + b) \times (10 + 1) \\&= (10a + b) \times 10 + (10a + b) \times 1 \\&= 100a + 10b + 10a + b \\&= 100a + 10a + 10b + b \\&= 100a + 10(a + b) + b\end{aligned}$$

3. **Read the final line out loud.** “ $100a + 10(a+b) + b$. What does this say in plain language?”

- $100a$ — the **hundreds** digit is a (the original first digit).
- $10(a+b)$ — the **tens** digit is $a + b$ (the digit sum).
- - b — the **units** digit is b (the original second digit).

“That IS the trick. The algebra spells out: first digit, sum, last digit.”

4. **Check on 23:**

$$a = 2, \quad b = 3.$$

$$100 \cdot 2 + 10 \cdot 5 + 3 = 200 + 50 + 3 = 253. \quad \checkmark$$

5. **Now the carry case** — where does the carry come from?

$$a = 8, \quad b = 7.$$

$$100 \cdot 8 + 10 \cdot 15 + 7 = 800 + 150 + 7 = 957.$$

“Notice: $10 \times 15 = 150 = 100 + 50$. The 100 contribution slides into the hundreds column — that IS the carry. So the algebra ALSO explains why we add 1 to the leftmost digit when the sum is ≥ 10 .”

Activity (15 min) — Explain to a friend

- Worksheet prompt:

Imagine a Grade 4 cousin asks: “*Why does the $\times 11$ trick work? Isn’t it just magic?*” Write 5–7 sentences explaining why it’s NOT magic. Use the words **distribute**, **digit sum**, and **place value**.

- 10 min individual writing.
- 5 min pair-sharing: read your paragraph to your partner; they tell you one part that’s clear and one part that’s unclear.

Wrap-up (5 min)

- Read aloud one strong paragraph (with permission).
- Preview Day 4: “*Tomorrow — extend the trick to 3-digit numbers. The algebra will scale.*”

Student-facing content

Why the $\times 11$ trick works

Write any 2-digit number as $10a + b$ — the first digit is worth 10, the second is worth 1.

Distribute the multiplication by $11 = 10 + 1$:

$$(10a + b) \times 11 = (10a + b) \times 10 + (10a + b) \times 1 = 100a + 10b + 10a + b = \mathbf{100a + 10(a + b) + b}.$$

In plain language: **hundreds** = a , **tens** = $(a + b)$, **units** = b . That’s the trick.

When $a + b \geq 10$, the “ $10(a+b)$ ” term overflows the tens place by exactly one hundred — that’s the carry.

It is not magic. It is the distributive law.

Homework

- Apply the proof to a number you choose. Pick any 2-digit number, write it as $10a + b$, distribute by 11, and show the final form. Show all 5 lines.

Differentiation

- **Tier up:** Prove the same identity for $\times 111$ (multiplication by 111). What’s the equivalent “split-and-add” pattern? Test it on 23×111 .
- **Tier down:** Don’t worry about the carry-case derivation. Get the no-carry version of the proof working.

Teacher notes

- This is the conceptual core of the module. Don’t rush.
- Students who can write the algebra but can’t *speak* it should be pushed to verbalise — “distribute” is the verb that matters here.

- The most common error in student paragraphs: confusing “the trick” with “the answer.” The trick is the algorithm; the algebra justifies the algorithm.
- About 20% of students will object: “*But this is just multiplication, not anything special.*” That’s the right reaction. Honour it: “*You’re right — and that’s the deeper point. The ‘special’ name (Ekādhikena Pūrveṇa) is a label; the math underneath is regular algebra.*”

Citation(s) used in this lesson

- (None required for the algebra itself — it is original derivation.)
- Optional teacher reference: Tirthaji (1965), sūtra 1, for the framing.

Day 4 — Three-Digit $\times 11$

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will compute any 3-digit $\times 11$ product using the running “neighbour-sum” pattern, including carry cases, and will complete Part A of the formative quiz.

Materials

- Whiteboard.
- Printed worksheet of 10 three-digit $\times 11$ problems.
- Formative quiz Part A (quizzes/formative.md).

Lesson flow

Warm-up (5 min)

- Neighbour-sum chorus: 10 quick 2-digit $\times 11$ problems including carry cases. (Should be near-instant by now.)

Core (18 min) — The 3-digit pattern

1. **Set up.** “Yesterday we proved $(10a+b)\times 11$. What happens with a 3-digit number $100a + 10b + c$?”
2. **Derive together** (or have students derive):

$$\begin{aligned}
 (100a + 10b + c) \times 11 &= (100a + 10b + c)(10 + 1) \\
 &= 1000a + 100b + 10c + 100a + 10b + c \\
 &= 1000a + 100(a + b) + 10(b + c) + c
 \end{aligned}$$

3. **Read the result.** “Thousands = a . Hundreds = $a + b$. Tens = $b + c$. Units = c .”

In words: **first digit stays, then each pair of adjacent digits sums and slides into the gap, then the last digit stays.**

4. **Worked example: 234×11 .**

Splits and gaps: 2 _ _ 4
 ↑ - - ↑
 (a+b) (b+c)

$$2 + 3 = 5$$

$$3 + 4 = 7$$

Answer: 2 5 7 4 → 2574.

Check: $234 \times 11 = 234 \times 10 + 234 = 2340 + 234 = 2574$. ✓

5. **Worked example with carry: 567×11 .**

5 _ _ 7
5 + 6 = 11 ← carry case
6 + 7 = 13 ← carry case

Strategy: work **right to left** to handle carries properly.

Units: 7.

Tens: $b + c = 6 + 7 = 13$. Write 3, carry 1.

Hundreds: $a + b + \text{carry} = 5 + 6 + 1 = 12$. Write 2, carry 1.

Thousands: $a + \text{carry} = 5 + 1 = 6$.

Answer: 6 2 3 7 → 6237.

Check: $567 \times 11 = 5670 + 567 = 6237$. ✓

6. **Generalisation.** For an n -digit number, the trick is the same pattern:

- Outer digits stay.
- Every interior digit becomes the sum of the two original digits flanking that position.
- Carries cascade from right to left.

Activity (15 min) — Worksheet

- 10 problems mixing 3-digit no-carry and carry cases:

123 × 11	234 × 11	345 × 11	456 × 11	142 × 11
365 × 11	478 × 11	529 × 11	687 × 11	818 × 11

- Pair work, self-check by long multiplication on any 3.

Formative Quiz Part A (5 min)

- Distribute quizzes/formative.md Part A.
- 5 quick problems, mixed 2- and 3-digit $\times 11$. Closed-notebook.

Wrap-up (2 min)

- Preview Day 5: “Tomorrow — speed day. We’ll retest your Day 1 problem (73×11) using the trick.”

Student-facing content

3-digit $\times 11$

For a number $abc \times 11$: 1. Write $a _ _ c$ (outer digits unchanged). 2. Fill the first gap with $a + b$. 3. Fill the second gap with $b + c$. 4. Handle carries right to left.

Example: $234 \times 11 \rightarrow 2 _ _ 4 \rightarrow 2 (2+3) (3+4) 4 \rightarrow 2574$.

The algebraic identity: $(100a + 10b + c) \times 11 = 1000a + 100(a+b) + 10(b+c) + c$.

Homework

- Solve 10 three-digit $\times 11$ problems on the back of today's worksheet. Time yourself — goal: under 90 seconds total.

Differentiation

- **Tier up:** Try a 4-digit $\times 11$ problem. Derive the pattern first (the identity will give you 5 terms). Test on 2345×11 .
- **Tier down:** Stay on 2-digit $\times 11$ problems with carries; build speed there before adding the 3-digit layer.

Teacher notes

- The right-to-left direction for carry handling is the key procedural insight. Some students will try left-to-right and run into trouble; redirect them.
- For 567×11 , the cascading carries (13 in tens, 12 in hundreds) are dramatic. Use that example.
- The pattern is sometimes taught with “underline carries” notation; we use right-to-left because it generalises better to n -digit numbers.

Citation(s) used in this lesson

- *Vedic_Maths.pdf* (ISKCON Desire Tree) — practice problems.
- Tirthaji (1965), sūtra 1.

Day 5 — Mental-Math Speed Day

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will retest their Day 1 baseline problem (73×11) using the trick, complete a paired speed race, and finish Part B of the formative quiz.

Materials

- Day 1 baseline sheet (returned to each student).
- Stopwatch.
- `activities/activity-01-speed-race.md` worksheet.

- Formative quiz Part B (quizzes/formative.md).

Lesson flow

Warm-up (3 min)

- Quick chorus: 12 problems, mixed 2- and 3-digit $\times 11$. Cover the carry case.

Day 1 retest (5 min)

- Return Day 1 baseline sheet to each student.
- Re-solve 73×11 using the trick. Time yourself.
- Record:

Day 1 time (long multiplication): _____ sec
 Day 5 time ($\times 11$ trick): _____ sec
 Improvement: _____ %

- Most students will see $>70\%$ reduction (e.g. 25 sec $\rightarrow$ 5 sec).

Activity (20 min) — Paired speed race

See activities/activity-01-speed-race.md for the full set-up.

- Round 1 (5 min): Solo, 20 mixed-difficulty problems. Each problem on a flashcard.
- Round 2 (5 min): Paired. Partner reads problem; you answer aloud.
- Round 3 (5 min): Group chorus — teacher displays 10 problems; class shouts the answer.
- Round 4 (5 min): Quick-fire — teacher calls a 2-digit number; first student to call back the $\times 11$ product gets a tally mark. (Light competition; no cumulative score across days.)

Formative Quiz Part B (15 min)

- Distribute Part B.
- 8 questions, mixed 2- and 3-digit $\times 11$, plus 2 short-answer “explain why” items.

Wrap-up (2 min)

- Preview Day 6: “*Tomorrow — the trick extends to $\times 12$, $\times 13$, ... all the way to $\times 19$. The middle-add step scales by a multiplier.*”

Student-facing content

Speed Day

Today’s question: how much faster has your mental $\times 11$ gotten in 4 days?

Record your Day 5 retest time. Compare with Day 1. The gap is the proof of practice.

By end of class you should hit roughly: - 2-digit $\times 11$ (no-carry): under 2 seconds - 2-digit $\times 11$ (carry): under 4 seconds - 3-digit $\times 11$: under 8 seconds

Homework

- 25-problem mixed drill sheet. Solo, untimed. Target: 100% correct.

Differentiation

- **Tier up:** Try 4-digit $\times 11$ problems. Sample: 1234×11 , 2345×11 , 4567×11 . Use the right-to-left carry-cascade method.
- **Tier down:** Focus only on the 2-digit drill today. Build fluency before adding 3-digit.

Teacher notes

- The Day 1 $\rightarrow$ Day 5 contrast is the emotional payoff of the module. Make sure every student records it; the personal evidence drives motivation.
- Quick-fire (Round 4) should NOT escalate into pure competition. Use it lightly; rotate “first responder” so no single student dominates.
- Some students will plateau on the carry case. Pair them with a peer-coach in Round 2 specifically.

Citation(s) used in this lesson

- (No new sources today — applied practice of Days 1–4.)

Day 6 — *Anurūpyeṇa*: Extending to $\times 12$, $\times 13$, ..., $\times 19$

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will apply Tirthaji’s *Anurūpyeṇa* sub-sūtra (“proportionally”) to multiply any 2-digit number by 12, 13, ..., 19, with at least 80% accuracy on a mixed worksheet.

Materials

- Whiteboard.
- `activities/activity-02-anurupyena.md` worksheet.
- Stopwatch.

Vocabulary introduced today

- *Anurūpyeṇa* (IAST: *anurūpyeṇa*; lit. “proportionally”) — a sub-sūtra in Tirthaji’s corpus. We use it as the name for the proportional scaling that extends the $\times 11$ pattern to $\times 12$, $\times 13$, ..., $\times 19$.

Lesson flow

Warm-up (5 min)

- Recall: state the $\times 11$ trick in one sentence.

- “Outer digits unchanged; middle is the digit sum.”

Core (18 min) — The scaling pattern

1. **Question.** “What does $\times 12$ look like? Try 23×12 by long multiplication first.”

$$23 \times 12 = 23 \times 10 + 23 \times 2 = 230 + 46 = 276.$$

2. **Derive.** “Now distribute $(10a + b) \times 12 = (10a + b)(10 + 2)$:

$$\begin{aligned} &= 100a + 10b + 20a + 2b \\ &= 100a + 10(b + 2a) + 2b \end{aligned}$$

Hmm — that’s awkward. $2b$ in the units, $(b + 2a)$ in the tens. Let’s organise it more cleanly.

3. **The Tirthaji pattern.** For N in $\{11, 12, \dots, 19\}$, write $N = 10 + k$ where $k \in \{1, 2, \dots, 9\}$. Then:

$$\begin{aligned} (10a + b) \times (10 + k) &= 100a + 10b \cdot k + 10ka + bk \\ &\quad \text{Wait — let's redo carefully:} \\ &= (10a + b) \times 10 + (10a + b) \times k \\ &= 100a + 10b + 10ak + bk \\ &= 100a + 10(b + ak) + bk \end{aligned}$$

Hmm — bk could be 2 digits. The pattern is: **scale each digit by k as you slide along.**

Easier form for students: For $\times N = \times(10 + k)$:

- Start with the leftmost digit.
- Multiply it by k , add the next digit, write the units of that sum (carry as needed).
- Continue right.
- The rightmost step: multiply the last digit by k .

This is the **Tirthaji form of the Anurūpyeṇa extension.**

4. **Worked example: 23×13 . ($k = 3$.)**

Set up: 0 2 3 (pad with 0 on left to track the carry-out)

Right to left:

- Units: last digit $\times k = 3 \times 3 = 9$. Write 9.
- Tens: middle digit $\times k + \text{next-right digit} = 2 \times 3 + 3 = 9$.

Write 9.

- Hundreds: leftmost digit $\times k + \text{next-right} = 0 \times 3 + 2 = 2$.

Wait — we want the LEFTMOST as just itself with the carry...

(Pause — this is getting confusing. Let’s use a different presentation.)

5. **Cleaner presentation for $\times 12$ to $\times 19$.**

For $N \times 12$ of a 2-digit number ab : simply use long multiplication mentally, but in the order:

- units of answer = $b \times 2$ (carry if ≥ 10)
- tens of answer = $a \times 2 + b$ (plus carry; carry if ≥ 10)
- hundreds of answer = a (plus carry)

For $N \times 13$: replace $\times 2$ with $\times 3$ everywhere.

General: **For $\times(10 + k)$, multiply each digit by k and add the digit to its right (the “running scan”).**

6. **Examples** (boxed):

23×12 :

units: $3 \cdot 2 = 6$
tens: $2 \cdot 2 + 3 = 7$
hundreds: 2 (no carry)
→ 276 ✓

23×13 :

units: $3 \cdot 3 = 9$
tens: $2 \cdot 3 + 3 = 9$
hundreds: 2
→ 299 ✓

45×13 :

units: $5 \cdot 3 = 15$ → write 5, carry 1
tens: $4 \cdot 3 + 5 + 1 = 18$ → write 8, carry 1
hundreds: $4 + 1 = 5$
→ 585 ✓

Check 45×13 : $45 \times 10 + 45 \times 3 = 450 + 135 = 585$. ✓

7. **The name.** “*Tirthaji (1965) calls this proportional scaling **Anurūpyeṇa** — ‘proportionally.’ The $\times 11$ case is the special case $k = 1$, where the scaling is trivial.*”

Activity (15 min) — Mixed drill

See activities/activity-02-anurupyena.md.

- 16 problems across $\times 12$, $\times 13$, $\times 14$, $\times 17$, $\times 19$.
- Paired self-check.

Wrap-up (2 min)

- “How does $\times 11$ fit into the $\times 12$ – $\times 19$ family?” (It’s the $k=1$ special case.)
- Preview Day 7: “What about $\times 23$? $\times 47$? When does the trick stop helping?”

Student-facing content

The $\times 12$ to $\times 19$ family — *Anurūpyeṇa*

For any $N = 10 + k$ (where k is 1 to 9), multiply a 2-digit number $ab \times N$ by: 1. Units = $b \times k$ (carry if ≥ 10). 2. Tens = $a \times k + b$ (plus any carry; carry if ≥ 10). 3. Hundreds = a (plus any carry).

When $k = 1$ (i.e. $\times 11$), this is the original “split and add” trick.

Tirthaji (1965) names this scaling pattern **Anurūpyeṇa** — “proportionally.”

Homework

- 12 problems mixing $\times 12$, $\times 13$, $\times 14$: see worksheet attached. Self-check with long multiplication on any 4.

Differentiation

- **Tier up:** Try 3-digit $\times 13$ problems. Use the right-to-left running-scan from Day 4 with the k -scaling from today.
- **Tier down:** Focus only on $\times 12$ today. Build fluency before adding higher k .

Teacher notes

- *Anurūpyeṇa* is the pattern-naming step. Some students will object: “*This is just long multiplication done right-to-left.*” They are correct. Honour the observation; reframe: “*The win is doing it MENTALLY in one pass, with a clear rhythm.*”
- The procedural complexity grows quickly with k . $\times 17$ and $\times 19$ are harder than $\times 11$ by a real margin. Calibrate the problem set accordingly.
- This day is genuinely harder than Days 1–5. If the class is shaky, slow down — skip the harder k values and revisit on Day 9.

Citation(s) used in this lesson

- Tirthaji (1965), sub-sūtra *Anurūpyeṇa*.
- *Microsoft Word - sutras1.pdf* — for the sub-sūtra list.

Day 7 — When the Trick Doesn’t Help

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will identify which multiplications are well-suited to the *Ekādhikena/Anurūpyeṇa* family and which are not, and will articulate the conditions that make a different sūtra (*Nikhilam* or *Ūrdhva-tiryagbhyāṁ*) more appropriate.

Materials

- Whiteboard.
- 8-problem comparison worksheet ($\times 11$ trick vs. long multiplication vs. *Nikhilam*-style).

Lesson flow

Warm-up (5 min)

- Quick chorus: 10 problems from Day 6's $\times 12$ – $\times 19$ family.

Core (20 min) — Where the trick breaks down

1. **Try $\times 23$.** “Let’s try the same proportional pattern with multiplier $23 = 20 + 3$.”
 - For 14×23 , would the trick scale?
 - $(14)(20 + 3) = 280 + 42 = 322$. Long way works.
 - The “proportional middle” pattern doesn’t extend cleanly past 19 because the multiplier has two non-trivial digits.
2. **Why?** Recall the identity: $\times(10 + k) \rightarrow 100a + 10(ak + b) + bk$. Clean because the “tens” part of the multiplier is **just 1**. For $\times 23$, the algebra is:

$$\begin{aligned}(10a + b)(20 + 3) \\&= 200a + 20b + 30a + 3b \\&= 200a + 30a + 20b + 3b \\&= (100 \cdot 2 + 100 \cdot 0 + \dots) - \text{no longer a tidy "outer-stays, inner-sums" pattern.}\end{aligned}$$

3. **The honest summary.**

The $\times 11$ trick (*Ekādhikena Pūrveṇa*) and its scaled cousins (*Anurūpyeṇa* for $\times 12$ to $\times 19$) work because the multiplier is *very close to 10*. Specifically, the multiplier must be $10 + k$ where k is a single digit.

For multipliers farther from 10 ($\times 23$, $\times 47$, $\times 98$), a DIFFERENT sūtra is the right tool — Tirthaji’s *Nikhilam* (“all from 9, last from 10”) for multipliers near 100, or *Ūrdhva-tiryagbhyāṁ* (“vertically and crosswise”) for general multiplication.

4. **Connect to Module 9 / 11.** Future modules will cover *Nikhilam* and *Ūrdhva-tiryagbhyāṁ* in depth. Today we just note their existence.
5. **The bigger lesson.** “Every mental-math shortcut has a domain. Knowing *WHEN* a trick applies is part of being a fluent mathematician. The standard long-multiplication algorithm always works; the shortcuts are faster only inside their domain.”

Activity (15 min) — Comparison drill

- 8 problems. For each, decide *before* solving: would you use (A) the $\times 11$ family trick, (B) long multiplication, or (C) a different mental shortcut you can articulate?

- | | |
|-------------------|--------------------|
| 1. 67×11 | 5. 84×23 |
| 2. 47×13 | 6. 99×99 |
| 3. 56×19 | 7. 25×12 |
| 4. 38×47 | 8. 102×11 |

- Pair work. Solve and discuss your method choice.

- Answers and “best method” given on the back.

Wrap-up (5 min)

- Class compiles “the $\times 11$ family domain”: *all 2-digit numbers $\times (11, 12, 13, \dots, 19)$ — about $90 \times 9 = 810$ distinct problems. That’s a significant mental-math territory.*
- Preview Day 8: “*Tomorrow — where do these 16 sūtras come from? The honest history.*”

Student-facing content

The domain of the $\times 11$ trick

The “split and add” trick works for multipliers of the form $10 + k$ where k is a single digit (so $\times 11$ through $\times 19$).

For other multipliers, use: - Long multiplication (always works). - *Nikhilam* “all from 9, last from 10” — when both numbers are near 100 (e.g. 98×97). - *Ūrdhva-tiryagbhyām* “vertically and crosswise” — general multiplication.

Knowing **when to use** a trick is as important as knowing **how**.

Homework

- Find any 3 multiplication problems in your homework or a textbook. For each, decide which method ($\times 11$ family / long / other) you’d use and **why**. Write 1 sentence of justification per problem.

Differentiation

- **Tier up:** Find any 2-digit $\times$ 2-digit multiplication where the $\times 11$ trick is NOT optimal but a different mental-math shortcut would beat long multiplication. (Examples: $25 \times 24 = 25 \times 24 = 600$ via $25 \times 4 \times 6$; $75 \times 12 = 900$ via $75 \times 12 = 75 \times 12$.)
- **Tier down:** Focus on identifying $\times 11$ vs. non- $\times 11$ cases. Skip the “best alternative method” question.

Teacher notes

- This day is short by design — it’s the honesty hour. Don’t over-fill.
- Some students will resist the framing “*the trick has a limited domain*” and try to force-fit $\times 23$ into a pattern. Honour the impulse, but redirect: “*You’d be inventing your own technique — which is great, but it’s not what this trick does.*”
- Mention that Tirthaji’s 16 sūtras collectively cover a wide territory; $\times 11$ is just one of them. This sets up Day 8.

Citation(s) used in this lesson

- Tirthaji (1965), sūtra list (especially *Nikhilam* — sūtra 2, and *Ūrdhva-tiryagbhyām* — sūtra 3).
- *Microsoft Word - sutras1.pdf* — full sūtra list.

Day 8 — Tirthaji's Sūtra System in Context

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will name at least 5 of Tirthaji's 16 sūtras, explain in 2–3 sentences when and by whom *Vedic Mathematics* was published, and state honestly that the “Vedic” attribution is academically contested.

Materials

- Printed handout: the list of 16 sūtras + 13 sub-sūtras (from Microsoft Word - sutras1.pdf).
- Optional: a paragraph excerpt from Tirthaji's preface (teacher's prepared synopsis).

Lesson flow

Warm-up (3 min)

- Recall: name the sūtra behind the $\times 11$ trick. (*Ekādhikena Pūrveṇa*.)
- Name the sub-sūtra behind $\times 12$ – $\times 19$. (*Anurūpyeṇa*.)

Core (25 min) — The 16 sūtras and the provenance question

- The 16 sūtras.** Hand out the list. Walk through abbreviated:

#	Sanskrit	English gloss
1	<i>Ekādhikena Pūrveṇa</i>	“By one more than the previous”
2	<i>Nikhilam Navataścaramaṁ Daśataḥ</i>	“All from 9 and the last from 10”
3	<i>Ūrdhva-tiryagbhyāṁ</i>	“Vertically and crosswise”
4	<i>Parāvartya Yojayet</i>	“Transpose and apply”
5	<i>Śūnyaṁ Sāmyasamuccaye</i>	“When the sum is the same, that sum is zero”
6	<i>(Ānurūpye) Śūnyam Anyat</i>	“If one is in ratio, the other is zero”
7	<i>Sankalana Vyavakalanābhyām</i>	“By addition and by subtraction”
8	<i>Pūraṇāpūraṇābhyām</i>	“By completing or non-completing”
9	<i>Calana-Kalanābhyām</i>	“By differentiation and integration”
10	<i>Yāvadūnam</i>	“Whatever the extent of its deficiency”
11	<i>Vyaṣṭi-Samaṣṭiḥ</i>	“Specific and general”
12	<i>Śeṣānyaṅkena Caramena</i>	“The remainders by the last digit”
13	<i>Sopāntya-dvayam Antyam</i>	“The penultimate and the last”
14	<i>Ekanyūnena Pūrveṇa</i>	“By one less than the previous”
15	<i>Guṇitasamuccayaḥ</i>	“The product of the sum is the sum of the products”

#	Sanskrit	English gloss
16	<i>Guṇaka Samuccayaḥ</i>	“All the multipliers”
Plus 13 sub-sūtras (including <i>Anurūpyeṇa</i>).		

2. Who was Tirthaji?

- **Bharati Krishna Tirthaji** (1884–1960). Śaṅkarācārya of Govardhana Maṭha (Puri) — i.e. head of one of the four major Hindu monastic seats founded by Ādi Śaṅkara.
- Studied at Madras Christian College. Multi-lingual scholar.
- Wrote the manuscript of *Vedic Mathematics* sometime between roughly 1911 and 1918 according to its editor’s foreword.
- The manuscript was lost or unfinished for decades.
- Tirthaji **died in 1960**.
- The book was **published in 1965**, five years after his death, edited by Vasudev Sharan Agrawala.

3. The provenance question. “Tirthaji claimed the 16 sūtras come from a Vedic text called the Atharva-veda Paṛiśiṣṭa — a supplement to the Atharva Veda.”

- **The problem:** No other Sanskritist has produced a manuscript of this *Paṛiśiṣṭa* containing these 16 sūtras.
- **The mathematician S. G. Dani** (TIFR, IIT Bombay) wrote a careful critique in 1993, reprinted in *Resonance* (October 2001), pointing out:
 - No manuscript evidence.
 - The Sanskrit style of the sūtras is more recent than Vedic.
 - The mathematical techniques have 19th–20th century parallels.

4. Honest framing. Two things can be true at once:

1. The $\times 11$ trick (and the rest of Tirthaji’s 16-sūtra system) **works**. It is a useful, beautiful piece of mental arithmetic.
2. The trick is **not verifiably ancient Vedic mathematics**. The best current evidence is that Tirthaji synthesised it in the early 20th century, drawing on classical Indian arithmetic + his own pedagogical genius, and named the sūtras in Sanskrit.

The synthesis itself is a real Indian intellectual achievement. The “Vedic” naming is a 20th-century interpretive choice that scholars dispute.

5. **What’s at stake.** If the sūtras are Vedic (3000+ years old), they’re a major historical claim. If they’re 20th-century synthesis, **they’re still a real achievement** — just a different one. Tirthaji’s pedagogical genius is not diminished by being recent.

Discussion (15 min) — Take a position

Pose two questions to the class. Take a brief show-of-hands; then take 4–5 student statements (no debate, just listening).

1. “Does it matter whether the trick is ‘ancient’ or ‘20th-century’? Why or why not?”
2. “If it turns out to be 20th-century, should we still teach it as ‘Vedic mathematics’? What should we call it?”

End by noting: “At Senior band you’ll read Dani’s critique in full and form a deeper position.”

Wrap-up (2 min)

- Preview Day 9: “Tomorrow — mixed-practice day plus project work. Friday is the final assessment.”

Student-facing content

Tirthaji and the 16 sūtras

Bharati Krishna Tirthaji (1884–1960), Śaṅkarācārya of Govardhana Maṭha, wrote *Vedic Mathematics* between roughly 1911 and 1918. The book was published in 1965, five years after his death.

Tirthaji claimed the 16 sūtras (including our *Ekādhikena Pūrveṇa*) come from a Vedic appendix-text. **No other Sanskritist has produced this manuscript.** S. G. Dani (1993) argues the sūtras are best understood as a 20th-century synthesis with Indian-mathematics roots, not a verifiably ancient Vedic teaching.

The trick works. The history is contested. Both can be true. The Indian intellectual achievement is real either way.

Homework

- Read the 16-sūtra list. Pick ONE sūtra (other than the two we’ve studied) and find what mathematical operation it’s used for. (Sūtras 2, 3, 4, 10, 14 are well-documented in Tirthaji 1965.) Write 3 sentences.

Differentiation

- **Tier up:** Read a one-paragraph excerpt from Dani (1993). What is Dani’s strongest critique? What does he NOT criticise?
- **Tier down:** Memorise the names of sūtras 1, 2, 3, and 14 (the four most commonly taught). Skip the historicity discussion details.

Teacher notes

- Some students (and their families) believe firmly that the trick is “ancient Vedic mathematics.” Honour the belief; don’t dismiss it. Frame your position as “the academic consensus is more sceptical, and you’ll read the evidence at Senior band.”
- Avoid civilisational chest-thumping (“ancient Indians knew this before everyone”). The Indian achievement here is the synthesis and clarity of Tirthaji’s pedagogical work; that’s enough to celebrate without overreach.
- Avoid the opposite error too — dismissing the trick as “just Western math with Sanskrit labels.” The pedagogical insight (the named patterns, the rhythm) is genuinely Tirthaji’s.

Citation(s) used in this lesson

- Tirthaji, B. K. (1965). *Vedic Mathematics*. Motilal Banarsidass. — Sūtra list and preface.
- Dani, S. G. (1993). “Myths and Reality: On Vedic Mathematics.” Reprinted in *Resonance* (October 2001).
- *Microsoft Word - sutras1.pdf* (SutraGanita corpus) — the 16+13 sūtra list.

Day 9 — Mixed Practice + Project Session

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will complete a mixed-practice drill covering 2-digit $\times 11$, 3-digit $\times 11$, and $\times 12$ – $\times 19$ problems, and will make substantial progress on their project (video, workbook, or proof poster).

Materials

- Mixed-practice drill sheet (20 problems).
- Project materials per pair: paper, markers, phone (for video pairs), template sheets (for workbook pairs).
- `assessments/rubric.md` posted on the wall for reference.

Lesson flow

Warm-up (3 min)

- 8 quick mental $\times 11$ problems including carry. Should be near-instant.

Mixed-practice drill (12 min)

- Distribute 20-problem sheet:
 - 5 problems: 2-digit $\times 11$ (no-carry)
 - 5 problems: 2-digit $\times 11$ (carry)
 - 5 problems: 3-digit $\times 11$
 - 5 problems: 2-digit $\times 12$ through $\times 19$
- Solo, untimed, but aim to finish in 10 min.
- 2 min: pair-swap, peer-check three problems each.

Project session (28 min)

- Project format options (pick on Day 8 or earlier):
 - **Format 1: 60-second teaching video** for a Grade 4 audience.
 - **Format 2: Printed workbook** — 25 problems organised in difficulty levels, with worked examples and an answer key.

- **Format 3: Proof poster** — one A3 sheet showing the algebraic derivation of $(10a + b) \times 11 = 100a + 10(a+b) + b$ with examples and a 3-line explanation.
- Each pair shows the teacher their plan in the first 5 min. Teacher signs off or redirects.
- Work time. Teacher circulates:
 - Check video pairs have a quiet recording space.
 - Check workbook pairs have grid paper.
 - Check poster pairs have A3 sheets and at least 2 markers.
- Final 5 min: each pair shares their working title and ONE thing they're stuck on. Class offers one suggestion.

Wrap-up (2 min)

- *“Tomorrow is the final assessment and project share. Bring your finished artefact. Polish, don't restart.”*

Student-facing content

Today's two jobs

1. **Drill.** 20 mixed problems. Self-assess.
2. **Project work.** Make real progress today — by end of class you should be 70% done.

Project Day 10: 2-minute presentation + 1-min Q&A. Be ready.

Homework

- Finish your project at home. 30 minutes max. If it's not finishable in 30 minutes, **simplify it** — better a complete simple project than a half-done ambitious one.

Differentiation

- **Tier up:** Add a “challenge” section to your project: a 4-digit $\times$ 11 worked example with cascading carries.
- **Tier down:** Use the workbook template provided. Don't custom-design from scratch today.

Teacher notes

- The project here is intentionally smaller-scope than Module 5's, because the technique is narrower (just $\times 11$ plus the $\times 12$ – $\times 19$ family) and the assessment is more on the algebra than on the artefact.
- Watch for pairs who haven't started by mid-class. Pull them aside; offer the simplest format (proof poster) as a fallback.
- The proof poster format is recommended for pairs that struggled on Day 3 — making the algebra public forces them to internalise it.

Citation(s) used in this lesson

- (No new citations — application day.)

Day 10 — Final Assessment + Project Showcase

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate mastery of the $\times 11$ family of techniques via a summative quiz and a 2-minute project presentation.

Materials

- Summative quiz (`quizzes/summative.md`).
- Stopwatch.
- Rubric (`assessments/rubric.md`) per pair.
- Reflection slip per student.

Lesson flow

Warm-up (2 min)

- “Quiz first (15 min), then presentations.”

Part 1: Summative Quiz (15 min)

- Distribute `quizzes/summative.md`. Closed notebook.
- Covers: 2-digit $\times 11$ (with and without carry), 3-digit $\times 11$, *Anurūpyeṇa* extension, the algebraic identity, the historicity question (1 short-answer).
- Collect at 15-min mark.

Part 2: Project Presentations (25 min)

- 2-min presentation + 1-min Q&A per pair (3 min total per pair).
- ~8 pairs fit in 25 min. Reserve next class for remaining pairs if needed.
- During each presentation:
 - Teacher marks rubric in real-time.
 - One assigned audience member asks one question.
 - Other audience members fill “I learned / I wonder” slips.

Wrap-up (3 min)

- Closing question (oral): “What’s one mental-math habit you’ll keep using after this module?”
- Reflection slip submitted (reproduced in `student-workbook.md`).

Student-facing content

Today: quiz + presentations

15 min for the summative quiz (closed book).

2-minute presentation per pair, then 1-min audience question.

Reflection (due at end of class): 1. What was the most surprising thing you learned about $\times 11$? 2. Did your view of “Vedic mathematics” change during this module? How? 3. If you had three more days, what would you do differently?

Homework

- None. Module is complete.

Differentiation

- **Tier up:** Students finishing the quiz early can attempt a bonus “challenge” sheet — 4-digit $\times 11$ plus a 2-digit $\times 19$ mix.
- **Tier down:** Open-notebook quiz for students with documented learning needs.

Teacher notes

- Mark the quiz immediately after class while the presentations are fresh.
- Don’t rush the reflection — the “did your view change?” question is the cultural-honesty payoff of the module.
- If presentations spill into Day 11, protect that time.

Citation(s) used in this lesson

- All sources verified across the module.

Module 7 (Middle) — Formative Quizzes

Two parts: Part A on Day 4, Part B on Day 5.

Part A — Day 4 (5 min, 5 problems)

Name: _____ **Class:** _____

Solve each using the $\times 11$ trick. Show the split-and-add steps if you want partial credit.

1. $24 \times 11 =$ _____

2. $35 \times 11 =$ _____

3. $73 \times 11 =$ _____

4. $95 \times 11 =$ _____

5. $123 \times 11 =$ _____

Part A answer key

Q	Answer	Notes
1	264	No-carry

Q	Answer	Notes
2	385	No-carry
3	803	Carry: $7+3=10 \rightarrow 0$ middle, carry 1 into 7 $\rightarrow$ 8
4	1045	Carry: $9+5=14 \rightarrow 4$ middle, carry 1 into 9 $\rightarrow$ 10 $\rightarrow$ cascade
5	1353	3-digit: 1 (1+2) (2+3) 3 $= 1353$

Total Part A: 5 marks (1 per problem).

Part B — Day 5 (10 min, 8 problems + 2 short answer)

Name: _____ **Class:** _____

1. (1 mark) $41 \times 11 =$ _____

2. (1 mark) $56 \times 11 =$ _____

3. (1 mark) $67 \times 11 =$ _____

4. (1 mark) $88 \times 11 =$ _____

5. (1 mark) $99 \times 11 =$ _____

6. (2 marks) $234 \times 11 =$ _____

7. (2 marks) $478 \times 11 =$ _____

8. (2 marks) $605 \times 11 =$ _____

9. (2 marks, short answer) Write the algebraic identity that proves the 2-digit $\times 11$ trick. (Use a and b .)

10. (2 marks, short answer) In one sentence, what happens when the digit sum ($a + b$) is 10 or more?

Part B answer key

Q	Answer	Marks
1	451	1
2	616	1 (carry: $5+6=11$, middle 1, carry

Q	Answer	Marks
		to 5 → 6)
3	737	1 (carry: $6+7=13$, middle 3, carry to 6 → 7)
4	968	1 (carry: $8+8=16$, middle 6, carry to 8 → 9)
5	1089	1 (cascading carry)
6	2574	2
7	5258	2 ($4+7=11$ carry, $7+8=15$ carry; right-to-left: units 8, tens 5, hundreds 2, thousands 5)
8	6655	2 ($6+0=6$, $0+5=5$; no carries)
9	$(10a + b) \times 11 = 100a + 10(a+b) + b$	2 (1 for left side, 1 for right side)
10	Write the units digit of the sum in the middle; add 1 (the tens digit of the sum) to the leftmost digit.	2 (1 for “units digit in middle”; 1 for “carry into leftmost”)

Total Part B: 14 marks.

What to do with the results

- Class average below 8/14 = reteach Day 3 (the algebra) and Day 4 (3-digit). Combine Day 6 with a re-drill.
- Individual below 5/14 = pull aside on Day 6 for 10-min reteaching of the carry case.
- Class average above 12/14 = proceed to *Anurūpyeṇa* (Day 6) at planned pace; consider tier-up problems for Day 9.

Module 7 (Middle) — Summative Quiz

Time: 15 min · **Marks:** 25 · **Closed notebook**

Instructions: Name + class at top. Show working on multi-digit problems if you want partial credit. No calculators.

Name: _____ **Class:** _____

Section A — 2-digit $\times$ 11 ($5 \times 1 = 5$ marks)

Solve mentally. Write only the answer.

- $32 \times 11 =$ _____
- $47 \times 11 =$ _____
- $68 \times 11 =$ _____

d) $85 \times 11 =$ _____

e) $99 \times 11 =$ _____

Section B — 3-digit $\times 11$ ($3 \times 2 = 6$ marks)

a) $123 \times 11 =$ _____

b) $456 \times 11 =$ _____

c) $789 \times 11 =$ _____

Section C — *Anurūpyeṇa* ($\times 12$ to $\times 19$) ($4 \times 1.5 = 6$ marks)

a) $23 \times 12 =$ _____

b) $34 \times 13 =$ _____

c) $45 \times 14 =$ _____

d) $27 \times 19 =$ _____

Section D — The algebra (3 marks)

6. Prove that $(10a + b) \times 11 = 100a + 10(a + b) + b$. Show all steps.

Section E — Open response (5 marks)

7. In 4–5 sentences, answer: > “*Is the $\times 11$ trick ‘ancient Vedic mathematics’?*” > > Use what you learned about Tirthaji, the date *Vedic Mathematics* was published, and at least one of the academic critiques.

Answer key

Q	Answer	Marks
A.a	352	1
A.b	517	1
A.c	748	1
A.d	935	1
A.e	1089	1
B.a	1353	2
B.b	5016	2 (4+5=9, 5+6=11 carry: units 6, tens 1, hundreds 0 + carry 1 = 1? Wait: $456 \times 11 = 4560 + 456 = 5016$. Right-to-left: units 6; tens 5+6=11 → 1, carry 1; hundreds 4+5+1=10 → 0, carry 1; thousands 4+1=5)
B.c	8679	2 ($789 \times 11 = 7890 + 789 = 8679$)
C.a	276	1.5
C.b	442	1.5 (34×13 : units $4 \cdot 3 = 12 \rightarrow 2$ carry 1; tens $3 \cdot 3 + 4 + 1 = 14 \rightarrow 4$ carry 1; hundreds $3 + 1 = 4$)
C.c	630	1.5 (45×14 : units $5 \cdot 4 = 20 \rightarrow 0$ carry 2; tens $4 \cdot 4 + 5 + 2 = 23 \rightarrow 3$ carry 2; hundreds $4 + 2 = 6$)
C.d	513	1.5 (27×19 : units $7 \cdot 9 = 63 \rightarrow 3$ carry 6; tens $2 \cdot 9 + 7 + 6 = 31 \rightarrow 1$ carry 3; hundreds $2 + 3 = 5$)
6	$(10a+b) \cdot 11 = (10a+b)(10+1) = 100a + 10b + 10a + b = 100a + 10(a+b) + b$.	3 (1 for distribution, 1 for rearrangement, 1 for final form)
7	A 4–5 sentence answer that: (a) acknowledges the trick was published in Tirthaji's <i>Vedic Mathematics</i> (1965, posthumous), (b) notes that Tirthaji claimed Vedic origin via <i>Atharva-veda Pariśiṣṭa</i> , (c) notes the academic critique (no manuscript; Dani 1993), (d) takes a clear position.	5 (2 for facts, 2 for citing a critique, 1 for clear position)

Total: 25 marks.

Grading bands (project combined → out of 50): - 40–50: Exceeds - 30–39: Meets - 20–29: Approaches - 0–19: Beginning

Homework

Homework — Module 7 Eleven Multiplication (Middle)

Day 1 → Day 2

Task: Find any five 2-digit numbers (with digit-sum < 10) around your house — phone numbers, page numbers, prices. Compute each $\times 11$ using the trick. Bring the worked list.

Time: 10 min.

Day 2 → Day 3

Task: Eight mental $\times 11$ problems including the carry case: 76, 84, 85, 88, 93, 96, 67, 78 — each $\times 11$. Write only the answers. Time yourself; target under 30 seconds total.

Time: 5 min.

Day 3 → Day 4

Task: Apply the proof to a number you choose. Pick any 2-digit number, write it as $10a + b$, distribute by 11, and show the final form. Show all 5 lines of working.

Time: 10 min.

Day 4 → Day 5

Task: Solve 10 three-digit $\times 11$ problems (on the back of today's worksheet). Time yourself — goal: under 90 seconds total. Self-check three with long multiplication.

Time: 15 min.

Day 5 → Day 6

Task: 25-problem mixed drill sheet (handed out in class). Solo, untimed. Target: 100% correct.

Time: 15 min.

Day 6 → Day 7

Task: 12 problems mixing $\times 12$, $\times 13$, $\times 14$ (worksheet attached). Self-check 4 with long multiplication.

Time: 15 min.

Day 7 → Day 8

Task: Find any 3 multiplication problems in your homework or a textbook. For each, decide which method ($\times 11$ family / long / other) you'd use and **why**. Write 1 sentence of justification per problem.

Time: 10 min.

Day 8 → Day 9

Task: Read the 16-sūtra list. Pick ONE sūtra (other than the two we've studied) and find what mathematical operation it's used for. Write 3 sentences. Confirm your project plan in writing (4–6 sentences).

Time: 20 min.

Day 9 → Day 10

Task: Finish your project. Polish, don't restart. Time your presentation in your head — must be 2 min.

Time: capped at 30 min.

Day 10 → end

Task: None. Module is complete. Optional: read ahead to Module 9 / 11 (Nikhilam, Ūrdhva-tiryagbhyāṁ) if curious.

Writing Prompts

Writing Prompts — Module 7 Eleven Multiplication (Middle)

Six prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

- 1. (Day 1 or Day 5, end of class, 3 min)** What did it feel like the first time you watched the teacher do 23×11 in 2 seconds? Describe what your mind did — disbelief, suspicion, “show me again”?
- 2. (Day 5 or Day 7, 5 min)** You've now seen one Vedic-math trick in detail. Has your *feeling* about mental math changed since Day 1? Be honest — even if the answer is “no.”

Explanatory prompts

- 3. (Day 3 or Day 4, homework, ~120 words)** Explain the $\times 11$ trick to a Grade 4 student who has never seen it. Use one specific example. No more than 6 sentences. Include the carry case in 1 sentence.
- 4. (Day 6 or homework, ~100 words)** Explain the *Anurūpyeṇa* extension in your own words. Why does the same idea scale from $\times 11$ to $\times 12$, $\times 13$, ..., $\times 19$? Use one example.

Argumentative prompt

5. (Day 8 or homework, 1 paragraph) Some people claim the $\times 11$ trick is “ancient Vedic mathematics.” Others say it’s a 20th-century pedagogical technique published in 1965. Pick a position and defend it in 6–8 sentences. Cite at least one source.

Creative prompt

6. (Day 4 or Day 9, optional, ~150 words) Write a short script (50–150 words) in which a frustrated student and a confident $\times 11$ -trick user have a conversation. The frustrated student raises one good objection (“this is just multiplication, why a special name?”). The confident user gives a clear, honest answer. Let the conversation be real — no “and then they magically agreed” endings.

Prompt-to-band notes

- For **Primary** band, prompts 1 and 6 work as oral dialogues; prompt 3 works as a drawing.
- For **Senior** band, prompt 5 deserves a full essay with citations from Tirthaji (1965), Dani (1993), and Plofker (2009).

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction.
- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the project rubric — does the writing acknowledge the other side, cite a source, reach a clear position?
- Prompt 6: no grading. Read the best ones aloud (with permission).

Activity 1 — Mental Math Sprint Cards

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Middle · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of $\times 11$ Multiplication — Ekādhikena Pūrveṇa through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

flashcards (20–50 pre-printed $\times 11$ problems), stopwatch

What students do

In pairs: one student holds up a card, the other answers within 3 seconds using the $\times 11$ method. Swap. Track personal best time.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 01 — Mental $\times 11$ Speed Race

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Day:** 5 · **Time:** 20 min

Type

Paired + group quick-fire drill

Goal

Each student does at least 20 mental $\times 11$ problems (mixed difficulty) in under 4 minutes, with peer-verified accuracy.

Materials per pair

- 1 printed flashcard deck of 24 problems (cut into separate cards if possible)
- 1 stopwatch
- Pen + paper for tally

Set-up (teacher, before class)

- Print and pre-cut the 24-problem flashcard deck. Mix the difficulty (8 no-carry, 8 carry, 8 three-digit).

Instructions (student-facing handout)

Speed Race — work in pairs across 4 rounds.

Round 1 (5 min, solo): Take 20 cards. Solve each silently in under 5 seconds. Self-check three random ones with long multiplication.

Round 2 (5 min, paired): Partner-A reads card 1 aloud. Partner-B answers in under 5 seconds. Swap roles. Continue until you’ve each done 10 problems out loud.

Round 3 (5 min, group chorus): Teacher displays a problem on the board. Class shouts the answer together. 10 problems total.

Round 4 (5 min, quick-fire): Teacher calls a 2-digit number. First student to call back the $\times 11$ product gets a tally mark. Light competition — rotate the “first responder” so no single student dominates.

Flashcard deck (24 problems)

Easy (no-carry, 2-digit, 8 cards):

12×11	23×11	34×11	52×11
61×11	16×11	27×11	35×11

Medium (carry, 2-digit, 8 cards):

47×11	58×11	69×11	76×11
83×11	89×11	97×11	99×11

Hard (3-digit, 8 cards):

123×11	234×11	345×11	456×11
142×11	365×11	478×11	529×11

Answer key (teacher copy)

Easy: 132, 253, 374, 572, 671, 176, 297, 385

Medium: 517, 638, 759, 836, 913, 979, 1067, 1089

Hard: 1353, 2574, 3795, 5016, 1562, 4015, 5258, 5819

Discussion prompts (last 2 min)

- Which type was hardest? (Most students say “carry” or “3-digit.”)
- Did anyone discover a faster sub-trick?
- Are you noticeably faster than Day 1? (Yes — name a number.)

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair alternating, one card per ~5 sec	“ $47 \times 11!$ ” — “ $517!$ ” — “ $58 \times 11!$ ” — “ $638!$ ”
Needs intervention	One partner stuck > 30 sec	Long silences; partner doing the math TO them, not WITH them

Safety / sensitivity

- Round 4 quick-fire risks dominating the room. If one student gets >4 tally marks while others have 0, switch to “must wait if you’ve answered the last 2.”
- No public leaderboards; tallies are private to each student.

Differentiation

- **Tier up:** Hard deck only. Try 4-digit $\times 11$ (sample: 1234, 2345, 4567).
- **Tier down:** Easy deck only. Build no-carry fluency before adding carries.

Clean-up

- Worksheets kept by students. Flashcards return to the bag for re-use next year.

Activity 02 — *Anurūpyeṇa* Drill ($\times 12$ to $\times 19$)

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Day:** 6 · **Time:** 20 min

Type

Solo + paired drill, mixed multipliers

Goal

Students mentally multiply 2-digit numbers by 12, 13, 14, 17, 19 using the *Anurūpyeṇa* pattern, with at least 80% accuracy.

Materials per pair

- 1 printed drill sheet (16 problems below)
- 1 stopwatch
- Pen each

Instructions (student-facing handout)

***Anurūpyeṇa* Drill** — work in pairs.

Round 1 (5 min, solo): Solve all 16 problems. No writing intermediate steps unless absolutely needed. Aim for the answer only.

Round 2 (5 min, paired): Partner-A reads problem 1. Partner-B answers within 10 seconds. Swap. Continue alternating through all 16.

Round 3 (5 min, mixed-multiplier): Teacher displays 8 new problems on the board, mixing all multipliers from $\times 11$ to $\times 19$. Class answers in chorus.

Round 4 (5 min, the hardest): 3 problems with $\times 19$ — the worst case. Show your working on paper this time.

Drill sheet (16 problems)

×12 problems:

1. 23×12 5. 41×12 9. 56×12 13. 78×12

×13 problems:

2. 24×13 6. 37×13 10. 45×13 14. 62×13

×14 problems:

3. 21×14 7. 35×14 11. 47×14 15. 53×14

×17 + ×19 (harder):

4. 16×17 8. 23×17 12. 27×19 16. 31×19

Answer key (teacher copy)

×12: 276, 492, 672, 936

×13: 312, 481, 585, 806

×14: 294, 490, 658, 742

×17 / ×19: 272, 391, 513, 589

Worked sanity check (give to slow pairs)

56×12 :

units = $6 \cdot 2 = 12 \rightarrow 2$, carry 1

tens = $5 \cdot 2 + 6 + 1 = 17 \rightarrow 7$, carry 1

hundreds = $5 + 1 = 6$

$\rightarrow 672 \checkmark$ (check: $56 \times 12 = 56 \times 10 + 56 \times 2 = 560 + 112 = 672$)

23×17 :

units = $3 \cdot 7 = 21 \rightarrow 1$, carry 2

tens = $2 \cdot 7 + 3 + 2 = 19 \rightarrow 9$, carry 1

hundreds = $2 + 1 = 3$

$\rightarrow 391 \checkmark$

Discussion prompts (last 2 min)

- Which multiplier was easiest? (Most students say ×12.) Why?
- Which was the hardest? (Most say ×19.) Why? (Bigger k means bigger carries.)
- How is the ×11 case the “easy” version of this same pattern?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair calmly working through carries on paper	“Units 12, carry 1; tens 17 plus 6 plus 1 is 24, carry 2...”
Needs intervention	Pair guessing answers; no working shown	“I don’t know, maybe 600?” — “Same.”

Safety / sensitivity

- Same as Activity 01 — no public leaderboards.
- Some students will hit a wall at $\times 17/\times 19$. That's expected — these are genuinely harder. Accept partial completion.

Differentiation

- **Tier up:** Try 3-digit $\times 13$ problems (e.g. 234×13). Same pattern, more digits.
- **Tier down:** Stick to $\times 12$ today. Build that fluency before adding $\times 13, \times 14$.

Clean-up

- Worksheets collected (teacher spot-checks before Day 7).

Project Brief

Project Brief — Module 7 Eleven Multiplication (Middle)

Work sessions: Day 9 · **Presentation:** Day 10 · **Total marks:** 35 (project 25 + presentation 10)

The project

Each pair (or individual, with teacher approval) chooses ONE of three formats:

Format 1: 60-second teaching video

Make a short video that teaches the $\times 11$ trick to a Grade 4 student. Constraints: - Total length: 60 seconds (± 10 sec). - Must show: at least 3 worked examples — one no-carry 2-digit, one carry case, one 3-digit. - Closing slide must cite Tirthaji (1965) AND a SutraGanita PDF.

Format 2: Printed workbook

Produce a 25-problem workbook with the $\times 11$ trick (plus 5 *Anurūpyeṇa* problems), designed for a Grade 4–5 self-learner. Constraints: - 25 problems organised in 4 difficulty levels: 7 no-carry 2-digit, 7 carry 2-digit, 6 three-digit, 5 *Anurūpyeṇa* ($\times 12$ – $\times 19$). - Each level includes ONE fully worked example. - Answer key on the back page. - Title page with cited source.

Format 3: Proof poster

Produce a one-page (A3) “proof poster” deriving $(10a + b) \times 11 = 100a + 10(a + b) + b$. Constraints: - The full derivation (at least 4 lines). - Three worked examples illustrating: no-carry, carry, cascading carry. - A 3–4 sentence “in plain English” explanation in your own words. - Bottom-strip citation of Tirthaji (1965) and the SutraGanita sūtra list PDF.

Why this matters

You're moving from *learner* to *teacher* (or *publisher*, or *prover*). Each role forces you to understand more deeply.

The proof poster is the format that requires you to *own* the algebra. We recommend it if you struggled on Day 3.

What success looks like

A strong project will: - Demonstrate the $\times 11$ trick (or its scaled cousins) accurately, no arithmetic errors. - Show the algebraic identity OR derive it in detail. - Be pitched for the intended audience. - Cite the sources properly. - Show your own thinking, not just rephrase classroom examples.

Working in pairs

- Pair work strongly encouraged.
- Individual allowed for the workbook or poster format.
- Groups of 3 acceptable — each member must have a distinct role.

Citation rules

- At minimum **one** citation must appear in your project — visible.
- Format: Tirthaji, *Vedic Mathematics* (1965), sūtra 1 OR <PDF filename>, <page or section> (e.g. *Microsoft Word - sutras1.pdf*).
- See `sources.md` for the list of available references.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to learn or make, and what did you actually end up with? 2. What was the most surprising thing you discovered while making this? 3. Did your view of “Vedic mathematics” change during this module? How?

Rubric

Project Rubric — Module 7 Eleven Multiplication (Middle)

Project marks: 25 (out of 35 with presentation 10).

Criteria

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Method demonstrated correctly	Trick shown completely (split, add, carry); at least 4 worked examples including 1	Trick shown completely; minor errors on 1 example.	Trick shown but with conceptual gap (e.g. carry case missing or wrong).	Trick incorrect or absent.	$\times 2$

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
	carry case; no errors.				
Algebraic justification	Includes the proof $(10a+b) \cdot 11 = 100a + 10(a+b) + b$, with at least 4 lines of derivation.	Includes the identity stated correctly.	Mentions the identity but no derivation.	Identity missing or incorrect.	$\times 1.5$
Use of sources	Cites Tirthaji (1965) AND a specific PDF section (e.g. <i>Microsoft Word - sutras1.pdf</i>).	Cites one source.	Mentions a source vaguely.	No source cited.	$\times 1$
Audience suitability	Pitched precisely to the target (Grade 4 for video; self-study for workbook; readable for poster).	Mostly suitable; minor tonal issues.	Suitable in places, off in others.	Inaccessible for target audience.	$\times 1$
Originality	The project shows the pair's own thinking — unique example, custom narrative, fresh framing.	Clearly the students' own work.	Largely repeated from class examples.	Copied verbatim from a source.	$\times 0.5$
Communication / craft	Visuals, layout, narration are clear, deliberate, polished. No errors in arithmetic shown.	Communicates clearly. 1–2 craft issues.	Message comes through with effort.	Hard to follow.	$\times 1$

Maximum raw = $(4+4+4+4+4+4) \times \text{respective weights} = 8+6+4+4+2+4 = 28$. Project score = $(\text{sum} \times \text{weights}) / 28 \times 25$.

Presentation (10 marks)

Criterion	4	3	2	1	Weight
Within time (2 min)	1:50– 2:00	1:40– 2:10	1:30– 2:20	Outside	$\times 1$
Both partners contribute	Equal	Both speak unequally	One dominates	One silent	$\times 1$
Clarity	Confident, audible	Audible, mostly clear	Audible only in front	Mostly inaudible	$\times 0.5$

Maximum raw = $4+4+2 = 10$.

Format-specific notes

- **Video projects:** the 60-second video itself is the artefact; the 2-min presentation explains the design choices (“we chose 87×11 because it’s the dramatic carry case”).
- **Workbook projects:** the workbook is the artefact; the 2-min presentation walks through the 2 hardest problems.
- **Proof poster projects:** the poster IS the artefact; the 2-min presentation IS the proof walked through aloud. This is the most algebra-heavy format.

What counts as “the algebraic justification”

We’re looking for the chain of derivation, not just the final identity. A complete answer shows:

$$\begin{aligned}
 (10a + b) \times 11 & \\
 &= (10a + b)(10 + 1) \\
 &= 100a + 10b + 10a + b \\
 &= 100a + 10(a+b) + b.
 \end{aligned}$$

Three intermediate steps. Each can be a line of working in the project.

Citation requirements

At least ONE citation must appear in your project — visible.

Format examples: - Tirthaji, **Vedic Mathematics** (1965), sūtra 1. -
Microsoft Word - sutras1.pdf – list of 16 Tirthaji sūtras.

For the proof-poster format, the citation can be a single line at the bottom; for video, in the closing credit; for workbook, on the title page.